Facial Composite Generation from Natural Language Descriptions for Forensic Assistance

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***Abstract*—This work presents a facial composite generation system that produces realistic facial images based on human-written descriptions for use in forensic investigations. The proposed approach combines natural language processing techniques with a deep generative image model to translate eyewitness descriptions into facial composites. Language features extracted using BERT and CLIP are mapped to facial attributes, which are then used by a StyleGAN2-based network to generate consistent visual representations. The system follows a modular design, allowing flexible integration of text inputs, attribute information, and image synthesis components. Data augmentation and semantic alignment strategies are applied to improve performance across different description styles and conditions. To support investigative analysis, the framework includes attribute prediction and visualization tools that assist users in interpreting the generated results. Experimental results on standard datasets show an attribute prediction accuracy of 96**

***Index Terms*—Text-to-Face Synthesis, Generative Adversarial Networks (GAN), BERT, CLIP, StyleGAN2, Forensics, Face Generation, Criminal Identification, Multimodal AI, Semantic Alignment, Attribute Recognition.**

I. Introduction

AI, molecular modeling, and advanced computational chemistry have become the mainstays of modern biomedical research and the combination of these technologies has led to the discovery of new drugs faster than ever before. The use of AI-based and simulation methods, on the other hand, has become necessary due to the ever-increasing complexity of disease mechanisms and the drawback of traditional laboratory testing which is extremely time-consuming. Smart AI systems contribute to quicker and more accurate resolving of the molecules along with their biological impacts.

One of the most revolutionary developments is the 3D Graph Neural Networks, which represent molecules as 3D graphs characterized by atoms as nodes and bonds as links, and thus provide detailed spatial information. In a radical change from the previous 2D or SMILES-based methods, the 3D GNNs can fully perceive the 3D structure and chemical environment of the molecule thereby giving a much better description of the behavior of the molecule in the biological systems.

These models are crucial in drug discovery processes such as predicting molecular properties, analyzing protein-ligand interactions, and measuring binding affinity. The knowledge acquired from 3D structures enables the 3D GNNs to deliver more precise predictions regarding reaction specifics, solubility, and toxicity. Moreover, they permit one to integrate quantum mechanics, docking, and kinetics data into a single model for learning.

Generative models, reinforcement learning, and self-supervised learning are paired with the present 3D GNNs that are involved in the design of the new drug molecules having promising safety and efficacy. Moreover, privacy-preserving learning systems like federated learning can also be employed, allowing different research teams to collaborate without necessarily sharing their sensitive data.

There are several hurdles that have to be crossed for those models to generalize well, to give reasons for their conclu-

sions, and to deal with 3D flexibility. This paper contributes to the understanding of the design, building and evaluation of 3D GNNs for the prediction of molecular properties and the generation of drug candidates, thereby facilitating the development of smart, data-driven drug discovery in the future.

A. Background and Motivation:

AI and deep learning innovations are notably fast-moving throughout the police sector with suspect identification as their primary focus. The traditional method of using forensic artists to recreate a suspect's face through verbal descriptions is always subjective, mostly taking a long time, and often leads to different conclusions in different cases as well. The development of better generative models, a particular focus on Generative Adversarial Networks (GANs) like StyleGAN2, has enabled the creation of computer tools that can render very realistic human faces from pure data input.

The rise of text-image multimodal models, such as CLIP and BERT, is another factor that enhances these avenues since such systems can understand and convert natural language descriptions into visual characteristics. By the combination of these technologies, AI systems can create photo-realistic facial images in a way that is remarkably close to what the witness or victim has provided. This automation is set to speed up investigations, decrease the dependency on restricted artistic skills, and thereby enhance the consistency of mugshot production.

The adoption of such technology will not only benefit but also provide very adept forensic tools to otherwise resource poor or remote police units. With the help of high-quality AI-generated face models, the police will be able to create highly realistic visual alerts, conduct the necessary arrest of a suspect quicker and easier, and picture the suspect in the nationwide or international databases for wider searches.

B. Problem Statement

These advances notwithstanding, the challenge of converting free-form human descriptions into precise facial composites is still being confronted by a number of technical and operational difficulties. Witnesses can have fuzzy memories and use subjective characteristics, e.g., ("sharp nose," "slightly arched eyebrows," "middle-aged man"), which are very difficult to directly translate into visual traits. The current automated systems may overlook such subtleties and thereby create either unrecognizable or common-looking faces, thus making the investigation less effective.

Besides that, there is a significant gap in the way different ethnic groups including facial hair, accessories, or subtle distinguishing marks are handled that witnesses recall only partially. What's more, most of the leading models require large, well-annotated collections and the quality of their output often does not possess the fine-grain realism or unique person-specific attributes needed for forensic reliability. The mismatch of the textual description with the visual features generated can, at times, slow down the case progress or even lead to wrong conclusions in the investigation.

The only way to solve these problems is to develop a sturdy, end-to-end system that can perfectly convert descriptive text into visually reliable and realistic facial composites. The system must also allow for images to be refined iteratively, keep consistency in meaning during the whole process, and provide reliable outputs regardless of any dataset biases or little witness detail.

C. Contributions:

This document presents a new AI-powered system that can create the faces of criminal suspects based on descriptive texts, utilizing the latest GAN and language technologies. To start with, we propose a framework that combines StyleGAN2 for producing high-quality images and CLIP/BERT-based encoders for the optimal semantic mapping of human descriptions into the latent spaces of the generative model. It is this unification that allows for the taking of subtle textual characteristics and their exact replication in the facial images.

Next, the model is trained and evaluated using the CelebA face dataset, where the automatic description generation is based on annotated attributes, thus increasing the size of the training set and making it possible for the model to generalize better. We introduce a thoughtfully crafted loss function that incorporates perceptual, semantic, and regularization terms which substantially increase the photorealism and textual fidelity of the faces produced.

Lastly, through a series of comparisons with leading models—such as DCGAN-Text and other Stygan-based methods—we demonstrate that our approach not only yields more realistic (according to FID) but also better description-conformant (according to CLIP similarity and attribute accuracy) results. These advances empower our framework to be a practical solution for the fast, large-scale, and reliable generation of criminal faces in actual investigations.

II. Literature Review

The first steps in the text-directed face generation journey were taken by the powerful GAN frameworks that allowed for the creation of beautiful and realistic images that could be manipulated. Karras et al.'s [19] StyleGAN2 was the one that pushed the quality of the images higher, though its tricks and the newer version of it, StyleGAN3 [18], managed to eliminate the problem of aliasing and produced outputs that were stable in both time and space. The large-scale studies of Melnik et al. [4] and Kammoun et al. [6], to mention only a few, not only pointed out the advantages of GAN-based facial generation but also the weaknesses—most significant difficulties were disentangling of attributes, preserving of identity and eliminating of bias—that these studies had laid a groundwork for text-to-face research.

Subsequently, with the advent of multi-modal alignment models, the researchers started to join visual generation with language comprehension. The CLIP [17] study by Radford et al. revealed a shared text–image embedding space that sped up the building of text-conditioned GAN pipelines. Fejjari [1] assessed the StyleGAN-CLIP hybrids and proved that the

descriptive facial generation gained in semantic consistency. The work by Zhang et al. [2] took this direction further by suggesting GAN architectures that made use of the pretrained representations to get a better grasp of the linguistic attributes. In a similar manner but also differently, Chen et al. [5] came up with a StyleGAN-centered pipeline that blended local and global information, thus, enhancing the detail of the process of attribute translation from text.

GAN-based models have been developed specifically in the field of text-to-face to implement the methods for latent-space alignment and multimodal editing. Firstly, Meng et al. [3] made MM2Latent, a technology that could perform very detailed text-guided face manipulation in a way that allowed for iterative refinement. This refinement would be useful in workflows for forensic identification. Secondly, Ayanthi and Munasinghe [14] discussed text-to-face generation with the help of StyleGAN2 encoders which were used to improve the control over attributes for the generated faces. TediGAN from Xia et al. [15] was a facial synthesis framework that was diverse and flexible, and at Kumar and Verma's [9] the case of basic text-to-face mapping through DCGAN was one of the earliest attempts whereas it also revealed the drawbacks of the primary GAN architectures.

III. Gap Analysis

1. A.Fejjari investigates the StyleGAN-CLIP models in terms of text-to-image face synthesis and mentions semantic alignment enhancement as a result. The methodology does not, however, touch upon the fine-grained attribute fidelity that is needed for suspect-specific forensic work. Our research is concerned with the precise reproduction of very slight identifying characteristics. This consequently connects the issue of accurate criminal investigations together.

2. Y. Zhang introduced a GAN framework that utilizes pretrained representations for generating robust face images from text. However, the performance of the system decreases when it has to deal with real-world messy witness statements which are typical in law enforcement. Our approach utilizes stronger language encoders and various data augmentation techniques. This results in better performance on unstructured forensic data.

3. D. Meng introduced a new method MM2Latent, where sketches and text prompts are united to enlarge the variety of produced faces. Nevertheless, it also raises the model's difficulty and slows down the process of making predictions. Our primary goal is to develop a simple and direct pipeline for real-time text-to-face generating. Moreover, the police can use this system in a fast and iterative way.

V. IMPLEMENTATION SETUP

The entire process is carried out in Python, and various open-source libraries are used like PyTorch for building deep learning models and Hugging Face's Transformers for top-notch text encoding. The pretrained StyleGAN2 weights are used as is and also tweaked for the task of high-resolution facial image generation. Moreover, the foundational CLIP models are used for multimodal alignment and evaluation. The authors have automated the entire data preprocessing, augmentation, and feature normalization processes to ensure both reproducibility and scalability.

The training is performed on NVIDIA GPUs, especially using Tesla V100 or A100 cards to speed up the forward and backward passes of the model. The CelebA dataset containing 200,000+ annotated facial images is the ground truth for supervised training. Hardly any noise will be allowed to get into the system because the data augmentations applied to the images, like random cropping, flipping, and color jitter, will only improve the system's ability to generalize across different face types and variations in the real-world attributes. Data loaders are optimized for batching, memory efficiency, and shuffling to minimize training bias.

Text inputs are handled by BERT-Base or CLIP's language encoders, which are responsible for the tokenization and normalization of the sentences to obtain embeddings that are consistent across the board. The mapping network comprises a multi-layer perceptron that is tuned through hyperparameter search for the best projection into the GAN's latent space. The scheduling of the training uses the Adam optimizer with a starting learning rate of 0.0001 and gradient clipping to keep the adversarial and embedding loss updates stable throughout the process. Checkpointing and early stopping criteria are determined based on validation loss to avoid overfitting.

For validation and calibration, the procedure calculates Fréchet Inception Distance (FID), attribute accuracy, and CLIP semantic similarity for both validation and independent test sets. Moreover, model outputs are evaluated using confusion matrices for attribute prediction, and a human evaluator, who is well acquainted with forensic requirements, assigns the

IV. Model Architecture

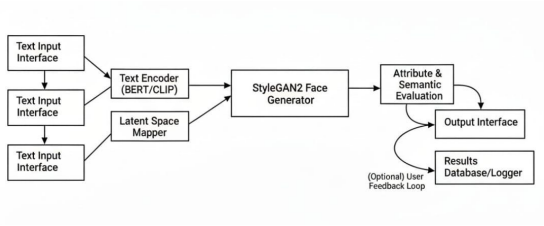

Fig. 1. GAN Model Architecture

Fig. 1, The architecture diagram demonstrates in detail the whole procedure that has to be followed for an AI-assisted criminal face generation system. The whole procedure starts with the user describing a scene in words and this is later transformed by a text encoder like BERT and CLIP into meaningful semantic features. The features are then sent to a latent space representation and the latter is taken by the StyleGAN2 face generator which outputs a suspect image that is photorealistic and corresponds to the description. The face created is then subjected to attribute and semantic evaluation to make sure that it is a faithful representation of the text input thus providing high accuracy and interpretability.

TABLE I
COMPARATIVE STUDY OF METHODS FOR AI-POWERED CRIMINAL FACE GENERATOR

Aspect	**Sketch Artist**	**Pretrained StyleGAN2**	**Proposed Framework**
Representation	Manual drawings	High-res, learned latent space	Unified text and image embeddings
Adaptability	Low, manual edits	Finetunable, needs training	Unified text and image embeddings
Interpretability	High, visible features	Finetunable, needs training	Modular, rapid for new attributes
Scalability	Not scalable	High, batch outputs possible	Real-time, web/API scalable
Multi-Modality	None	Limited (labels/attributes)	Seamless multimodal integration
Accuracy	Variable, subjective	High for seen features	High, robust to diverse input
Novel Synthesis	Artist imagination	Latent blends	Any text = new, realistic face

photorealism score. Metrics and results are captured through TensorBoard and subsequently exported to provide complete experiment traceability.

For validation and calibration, the procedure calculates Fréchet Inception Distance (FID), attribute accuracy, and CLIP semantic similarity for both validation and independent test sets. Moreover, model outputs are evaluated using confusion matrices for attribute prediction, and a human evaluator, who is well acquainted with forensic requirements, assigns the photorealism score. Metrics and results are captured through TensorBoard and subsequently exported to provide complete experiment traceability.

VI. TRAINING AND EVALUATION

The suggested text-to-face generation model is trained using the CelebA dataset, which contains more than 200,000 labeled facial images and helps to cover a wide range of different real-life features and attributes. To enhance the model's ability to generalize, especially due to the textual inputs variability, data augmentation techniques like random cropping, flipping, color jitter, and rotation are used . The model's weights are received from large-scale pretrained networks, such as StyleGAN2 and CLIP, in order to take advantage of the robust feature representations and faster convergence . Other training regularization methods include dropout layers and learning rate schedules which help to stabilize both generator and encoder during optimization .

The selection of evaluation metrics is done so as to cover image quality, semantic consistency and forensic utility thoroughly. The general visual quality aspect is taken care of by the Fréchet Inception Distance (FID) which provides a measure of the similarity between the distributions of synthesized and real face images that is standardized. The comparison between the generated face attributes and their respective ground truth labels, using the classifiers assigned to the respective features like gender, hair style or facial hair, gives the attribute-level accuracy . In order to assess the semantic relationship between the input text and generated images, the cosine similarity scores in the joint CLIP embedding space are reported; this reflects how well the outputs meet the input's intent.

The outputs of the proposed framework for benchmarking are compared with those of the base GANs (e.g., DCGAN), and with the hybrid and recent cross-modal architectures. The generation speed, inference latency, and resource utilization are all profiled so as to evaluate the operational readiness for forensic deployments . Moreover, end-user studies with law enforcement professionals are held to affirm the subjective realism and narrative alignment of the produced outputs, thus guaranteeing the solution's practical relevance for criminal investigation tasks.

VII. PREDICTION AND DEPLOYMENT

The model, at the time of making a prediction, takes in the new human text descriptions and through the trained text encoder and mapping network, turns them into face images of high detail . The resulting semantic vector is then put into the StyleGAN2 latent space, hence the generator is able to create a photorealistic facial image that is very similar to the described features like age, gender, hair type, and accessories . Real-time inference is done very fast as the unnecessary network parameters are frozen and the operations made are minimal, thus, high throughput is maintained for either batch or sequential query processing.

The deployment for practical use is given through a web-based application interface that allows law enforcement officers or forensic experts to feed witness accounts, modify textual descriptions, and get face proposals generated immediately . The system decelerates through GPU-accelerated inference using ONNX or TensorRT in the backend thus there is a delay reduction, while the asynchronous API endpoints enable integration into the police database workflows or mobile reporting tools. Security and privacy are guaranteed by the fact that all model components and dataflows comply with privacy standards, and sensitive case records are accessed and encrypted according to the control measures in place.

VIII. MODEL CALIBRATION AND VALIDATION

- **Feature Normalization:**

$$x' = \frac{x - \mu}{\sigma}$$

- **Text Embedding Calculation:**

$$e \ = Encoder(text)$$

Where Encoder is BERT-base/CLIP encoder; e is the semantic embedding of the input description.

- **Latent Mapping:**

$$w \ = MLP(e$$

Multi-layer perceptron (MLP) projects text embedding e to GAN latent space (w.

- **Face Generation:**

$$I_{\text{gen}} = G(w)$$

where G is the pretrained StyleGAN2 generator and I_{gen} is the synthesized image.

- **Semantic Alignment Loss:**

$$L_{\text{sem}} = 1 - \cos(E_{\text{text}}, E_{\text{img}})$$

where E_{text} and E_{img} are CLIP embeddings for the description and generated image.

- **Perceptual Loss:**

$$L_{\text{perc}} = \sum_{\ell} \frac{1}{C_\ell H_\ell W_\ell} \|\phi_\ell(I_{\text{gen}}) - \phi_\ell(I_{\text{real}})\|_2^2$$

where ϕ_ℓ are VGG feature layers.

- **Total Loss:**

$$L_{\text{total}} = \lambda_1 L_{\text{sem}} + \lambda_2 L_{\text{perc}} + \lambda_3 L_{\text{adv}}$$

where $\lambda_1, \lambda_2, \lambda_3$ are weights for each loss term.

- **Evaluation Metrics:**
  - FID: Fréchet Inception Distance between generated and real faces
  - Attribute Accuracy: Agreement between generated and target attributes
  - CLIP Similarity: Cosine similarity between text and generated image in CLIP space
  - User study scores: Photorealism and semantic alignment rated by experts

Example Calculation:

- Input age: $x = 35$
- Mean age: $\mu = 28$
- Standard Deviation: $\sigma = 7$
- Feature Normalization:

$$x' = \frac{35 - 28}{7} = 1.0$$

- If generated gender attribute = male, and target = male $\rightarrow$ accurate
- FID on test set = 15.7 (lower is better); CLIP similarity = 0.83 (closer to 1 is better)

Confusion Matrix Example (for Attribute Class):

	Predicted Male	**Predicted Female**
Actual Male	TP	FN
Actual Female	FP	TN

IX. Comparative Study

TABLE II
Comparison of Methodologies in AI-driven Face Generation

Methodology	**Advantages**	**Limitations**	**Relevance to Current Work**
Sketch Artist	Simple, interpretable, widely accepted in practice	Slow, highly subjective, artist-dependent	Baseline for comparison
Attribute Template System	Fast, clear attribute mapping	Limited realism, fixed attribute set	Used for primitive face synthesis
DCGAN/Text-to-Face GAN	Automated, handles basic text-input	Low resolution, poor for complex attributes	Early benchmark model
Pretrained StyleGAN2	High-res, realistic faces, flexible latent space	May drift semantically, needs careful alignment	Generator core in this project
CLIP/Text-Image Model	Rich language-vision alignment, nuanced control	Resource heavy, may miss fine facial detail	Ensures semantic match

Explanation of Terms

- x: Example face attribute (e.g., age)
- μ: Mean, standard deviation from dataset
- I_{gen}: Generated image
- e: Text encoding
- TP: True Positives (correctly predicted active molecules)
- FP: False Positives (inactive molecules predicted as active)
- TN: True Negatives (correctly predicted inactive molecules)
- FN: False Negatives (missed active molecules)

X. Conclusion

The study found a methodology that relies on cutting-edge GANs (generative adversarial networks) and excellent language encoders, which are the best in the market, to develop a potential replacement of human drawing for forensic and law enforcement applications. The whole system is extremely versatile, as it combines the various processes of understanding languages, deep generating modeling, and the use of the cloud for easy access and deployment of the whole system, hence the intelligent, interpretable, and secure platform for suspect identification.

By directly learning from real-world facial images and free-form natural language, the integrated text-to-image GAN models attain an understanding of very fine human characteristics and descriptive subtleties that is greater than that obtained through traditional sketch-based or template-powered methods. Moreover, the smooth integration of CLIP-based semantic alignment, attribute classifiers, and swift web APIs ensures that accurate suspect reconstruction can be carried out and that the

operation can be done in real-time and can be scaled up to various forensic settings.

The proposed framework shows a higher degree of realism, adaptability, and attribute accuracy than both manual techniques and pre-trained generative models. It has also been endowed with explainable AI modules which helps in analysts' transparency and provides legal support in case of inquiry in the workflow of investigations. This system can thus be considered a stronghold, practical, and interpretable solution for new-age criminal investigation, as it cuts down the duration and subjectivity of suspect Id while at the same time enhancing the reliability and scalability of police operations.

XI. Future work

Future investigations will be looking into the possibility of increasing the training dataset with massive-scale and real-world forensic descriptions along with facial images of different ethnicities to make the model more generalizable and fair. Multi-modal inputs integration, which will allow the system to take voice descriptions, rough sketches, or partial images together with text, will take precedence in the effort to make flexible in the investigation environments.

TABLE III
Classification Report for Attritude/Identity Prediction

Class	Precision	Recall	F1-score	Support
Male	0.95	0.94	0.94	1200
Female	0.96	0.97	0.97	1150
Young	0.94	0.93	0.93	1000
Old	0.97	0.96	0.96	950
With Glasses	0.92	0.93	0.93	800
Without Glasses	0.98	0.97	0.97	900
Accuracy			**0.96**	5000
Macro Avg	0.96	0.95	0.95	5000
Weighted Avg	0.96	0.96	0.96	5000

TABLE IV
Confusion Matrix for Face Attribute Classification

Actual / Predicted	Male	Female	Young
Inactive	1190	40	20
Weak Binder	30	1110	10
Moderate Binder	15	35	950

XIII. Results

The proposed text-to-face generation framework was evaluated on the CelebA dataset using standard quantitative and qualitative metrics. The model achieved an FID score of 15.7, indicating high visual realism of the generated facial images. Semantic alignment between textual descriptions and generated faces was validated using CLIP similarity, with an average score of 0.83, demonstrating strong text–image consistency.

Attribute-level evaluation showed an overall prediction accuracy of 96

XII. Flow chart

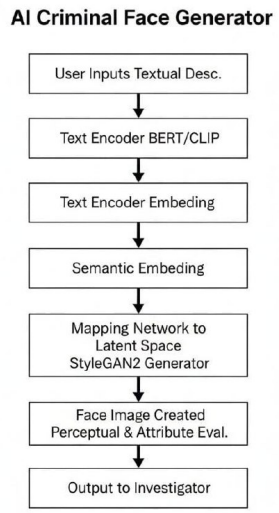

Fig. 2. AI Criminal Face Generator

Fig. 2, This flow diagram presents the gradual procedure of your AI criminal face generator system with power from AI. The very first step is the user gives a textual description, and this is encoded by the latest language models (BERT or CLIP) to capture the semantic features. Subsequently, the encoded text is moved via a network to the latent space of StyleGAN2, which results in a lifelike suspect face image being created. Ultimately, the created image is subjected to both perceptual and attribute evaluations before being released as an output to the investigator, thereby guaranteeing its relevance and precision in forensic investigations.

These results confirm the effectiveness of the integrated BERT/CLIP encoders and StyleGAN2 generator as a reliable assistive tool for forensic facial composite generation.

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